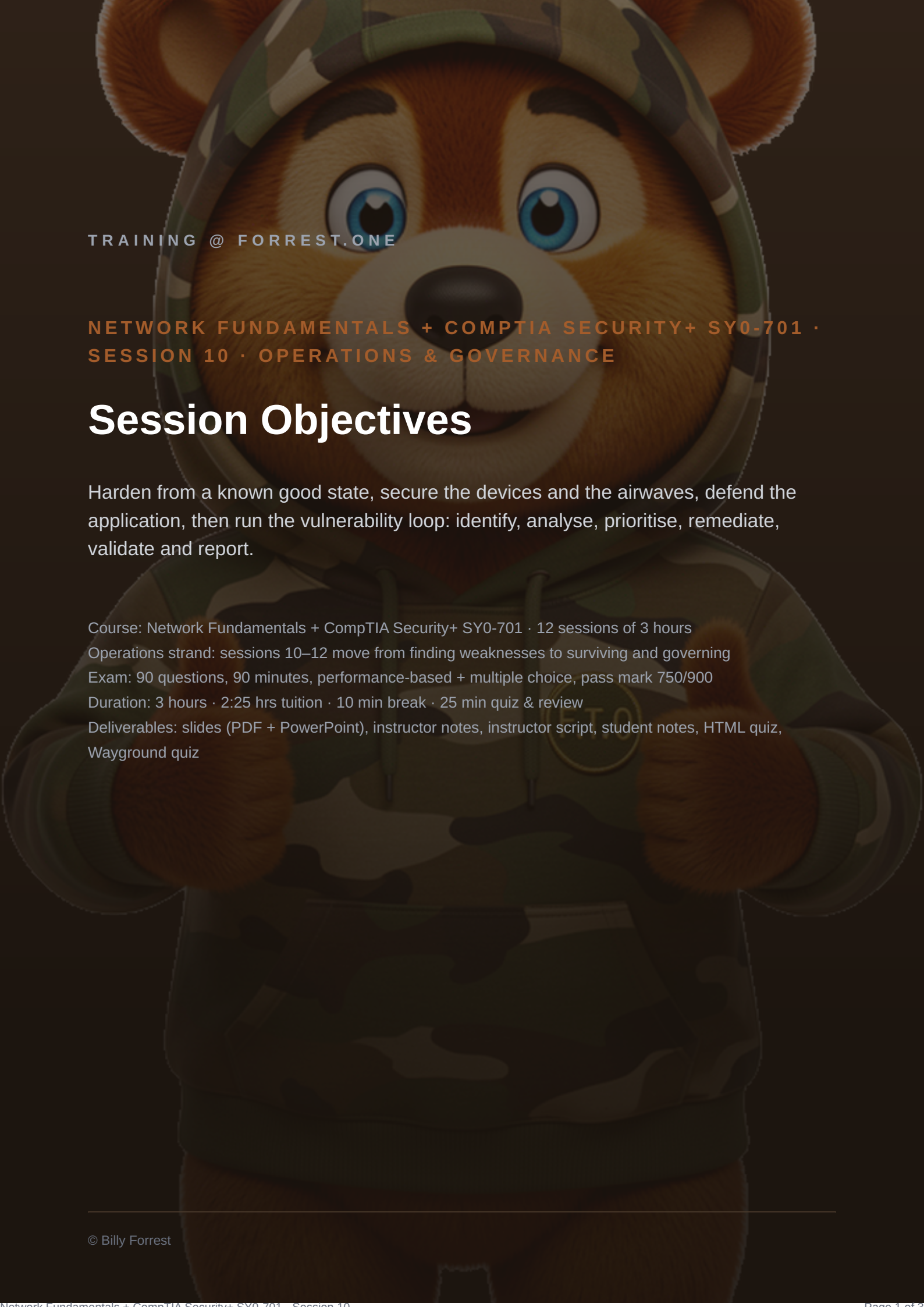




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 10 · OPERATIONS & GOVERNANCE

Session Objectives

Harden from a known good state, secure the devices and the airwaves, defend the application, then run the vulnerability loop: identify, analyse, prioritise, remediate, validate and report.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Operations strand: sessions 10–12 move from finding weaknesses to surviving and governing

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives: Vulnerability Management, Resiliency & Asset Protection

This session follows the expanded six-chapter spine: build from a secure baseline and hold the configuration there, secure the mobile devices that leave the building and the wireless medium they use, harden applications and monitor them, manage every asset from purchase to certified destruction, then run the vulnerability management loop of identify, analyse, prioritise, remediate, validate and report. It closes with the monitoring tooling that turns raw events into decisions.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Explain what a secure baseline is, and recognise configuration drift when it appears.
- Describe the four steps of implementing a baseline, and name the sources that inform it.
- Compare configuration management tools: Ansible, Puppet, Chef and SaltStack.
- Explain hardening as attack-surface reduction: services, ports, default accounts and least privilege.
- Apply hardening at three layers: operating system, application and firmware including secure boot.
- Balance security against functionality, and explain why over-hardening gets rolled back.
- Describe the unique risks of mobile devices and what mobile device management answers.
- Compare BYOD, corporate owned devices, containerisation and application vetting.
- Explain the four generations of wireless encryption and what WPA3 adds.
- Compare pre-shared key and enterprise wireless authentication, and detect rogue access points.
- Apply secure coding practice: least privilege, error handling, code review, input validation and boundary checking.
- Explain the application defences: web application firewall, sandboxing, allow listing, code signing and runtime self-protection.
- Describe security monitoring: what to watch, and how alerting strategy interacts with scope and storage.
- Manage the asset lifecycle: acquisition requirements, owner and custodian roles, tracking, and secure disposal.
- Explain data sanitisation methods, verification and chain of custody, including the regulatory duties.
- Run the vulnerability loop: identification, analysis, prioritisation, remediation, validation and reporting.
- Interpret CVSS base, temporal and environmental metric groups, and add context to decide priority.
- Describe the monitoring tooling: SIEM, log management, traffic analysis, file integrity monitoring, behaviour analytics and alert tuning.

Course context

Item	Detail
Course	Network Fundamentals + CompTIA Security+ SY0-701, 12 sessions of 3 hours
Strand	Operations and governance (sessions 10, 11 and 12)
Prerequisites	Sessions 1 to 9, the foundation and architecture strands
Assessment	HTML learning quiz (20 questions) + Wayground quiz (15 questions)
Exam link	Domain 4 operations: vulnerability management, monitoring, asset protection

Session 10 coverage

Chapter	Area	What is covered
1	Baselines and hardening	Secure baselines, drift, configuration management tools, hardening at three layers
2	Mobile and wireless	Mobile risk, BYOD and containerisation, wireless encryption, authentication, rogue access points
3	Applications and monitoring	Secure coding, input validation, application defences, monitoring fundamentals and alerting
4	The asset lifecycle	Acquisition, owner and custodian, tracking, secure disposal and chain of custody
5	Vulnerability management	Identification, CVSS analysis, prioritisation, remediation, validation, reporting
6	Monitoring tools	SIEM, log management, traffic analysis, file integrity monitoring, behaviour analytics

Course position and cross references

Resiliency, recovery objectives and backup strategy now sit in Session 7, and physical and environmental site controls in Session 8; this session keeps the asset and disposal controls that complete that story. Session 11 takes the evidence these controls produce into security operations, incident response and digital forensics. Every quiz question is tagged with the session area it tests (10A Identification through 10F Site Security); the slides use the book's subtopic codes (10A Baselines through 10L Tooling), and the instructor notes carry the mapping.